Manisha Muduli

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Summary

Photonics Engineer with 7+ years of experience in semiconductor process development, materials integration, and device fabrication. Expertise in thin-film processing, materials characterization, and III–V/Si heterogeneous integration, including 300 mm wafer processing and 3D integration. Skilled in DOE, TCAD, and correlating process and material properties with device performance.

Core expertise: Materials Integration • Thin-Film Processing • 300 mm Wafer Processing • Wafer Bonding • Materials Characterization • Interface Engineering • ALD/PVD/CVD • XPS • SEM/AFM • Ellipsometry • DOE • TCAD • JMP/MATLAB

Experiences

QC82 Inc. Charlottesville, VA
Photonics Scientist Aug 2026 – Present

- **Develop and optimize fabrication processes** for InGaAs-based SPADs targeting 1550 nm sensing applications.
- **Interface with clients and external vendors** on process design and integration requirements and technical deliverables.
- Establish device characterization capabilities through equipment selection and measurement-system planning.

Intel Corporation Hillsboro, OR
Research Intern, 3D Semiconductor Integration Technology May 2025 – Aug 2025

- Executed 300 mm **wafer bonding** & layer transfer processes in a **Class 1 cleanroom** for 3D heterogeneous integration.
- **Characterized and analyzed** bond integrity via **defect inspection**, interface analysis, and bond energy measurements.
- **Supported Intel–supplier discussions** by analyzing & organizing technical data, driving improvements in supplier tools.
- Presented data-driven results to Intel research leadership, contributing to next-phase development strategy.

The Ohio State University Columbus, OH
Graduate Research Associate / Research Assistant May 2021 – Aug 2026

- Developed III–V/Si heterogeneous integration flows using transfer printing and bonding for **SWIR photonic devices**.
- Developed lithography, thin-film deposition, etching, and metallization processes for III–V/Si integration.
- Applied **DOE, TCAD, and materials characterization** for process optimization and device analysis.
- **Coordinated cross-functional research** and proposal development with internal and external collaborators.

University of Pennsylvania Philadelphia, PA
Graduate Student Fellow / Research Assistant Jan 2019 – Apr 2021

- Executed full microfabrication process flows using PVD/CVD, lithography, and thin-film metrology in a cleanroom environment.
- Synthesized size-tunable InAs quantum dots and integrated them into spin-coated InAs/Si photodetector for SWIR.
- **Characterized quantum dots** using UV–Vis spectroscopy to validate bandgap control and material quality.
- Fabricated and electrically characterized InAs-on-Si photodetectors with varying absorber thicknesses.

Skills

- **Modelling & Data Analysis:** Silvaco TCAD, NovaTCAD, MATLAB, JMP, DOE, SRIM, Origin
- **Process & Integration:** Wafer Bonding, 3D Heterogeneous Integration, Micro-Transfer Printing, 300 mm Wafer Processing, Process Flow Development
- **Fabrication:** Photolithography, E-Beam Lithography, ALD, CVD, PVD, RIE, Wet/Dry Etching, Metallization, RTA
- **Characterization**: XPS, SEM, AFM, Ellipsometry, Profilometry, IV/CV, QE, PL, Optical Spectroscopy

Selected Technical Projects

GaAsSb/Si SACM Avalanche Photodiodes (1.55 µm) Jul 2024 - Aug 2026

- Optimized GaAsSb/Si SACM structures using **TCAD** to control electric-field distribution and carrier transport.
- Determined GaAsSb/Si band alignment via **XPS**, identifying < 0.1 **eV conduction-band offset**.
- Correlated interface/process conditions with electrical device behavior using **IV/CV characterization**.

Heterogeneous III–V/Si SWIR Photodetectors Dec 2023 - Apr 2025

- Developed **transfer-printing workflows** to integrate III–V photodetectors with silicon.
- Demonstrated an **electrically active III–V/Si interface and 1.55 μm photoresponse**.

Selective Etchant Studies for III–V Materials Dec 2023- Mar 2024

- Investigated **selective wet etchants** for controlled III–V layer removal and process integration.
- Optimized etch conditions using **selectivity, etch rate, and surface morphology** for transfer-printing workflows.

Zn Diffusion in InGaAs/GaAsSb Superlattices Jun 2022 - Jan 2024

- Varied **ALD ZnO thickness, temperature, and diffusion time** to control Zn incorporation in InGaAs/GaAsSb superlattices.
- Used **SIMS, IV, and CV** to confirm **PN-junction formation** and identify lateral diffusion for further optimization.

InAs Quantum Dot Photodetectors Jan 2019 - May 2020

- Designed experiments to control InAs QD size by varying **precursor and reducing-agent ratios**.
- Characterized QD optical properties using **UV–Vis spectroscopy**, correlating **QD size with bandgap**.

Leadership and Professional Involvement

- **SPIE Student Chapter**, Vice Present (2024-2026), Organized and led seminars, and managed event logistics.
- **Graduate Student and Postdoc Advisory Council**, ECE Delegate (2024-2026)
- **Mentorship**, Mentored 5+ students (highschool, master's and PhD) on research, device fab, and measurements.
- **Training Lead**,
  - **Led cleanroom training** and tool protocol enforcement to enhance lab operational reliability.
  - **Trained 10+ users** to ensure tool & user safety; **maintained & calibrated** cryogenic probe station.

Selected Publications

- **Muduli, M.**, et al., *SPIE Optical Interconnects & Packaging*, 2025 – Heterogeneous integration of InGaAs/GaAsSb superlattices with Si for e-SWIR sensors.
- **Muduli, M.**, et al., *Applied Physics Letters*, 2024 – Active interface characteristics of GaAsSb/Si photodiodes.
- Kodati, S., & **Muduli, M.**, *IntechOpen*, 2025 – Electrostatics in engineered photodiodes for Earth observation cameras.
- Xia, Y., **Muduli, M.**, et al., *Journal of Applied Physics* – Surface treatment of GaAsSb/Si via IV/CV analysis.
- Lee, J., et al., **Muduli, M.**, *Journal of Chemical Physics*, 2024 – One-pot synthesis of SWIR InAs quantum dots.
- Pise, M., **Muduli, M.**, et al., *Surfaces & Interfaces*, 2022 – Palladium nucleation and growth modeling.

Selected Conferences

- **SPIE Photonics West** - San Francisco, CA (2025), Oral Presentation.
- **Electronic Materials Conference** - College Park, MD (2024), Oral Presentation.
- **SPIE Defense + Commercial Sensing** - Orlando, FL (2023), Oral Presentation.

Education

The Ohio State University, OH, USA
Ph.D. in Electrical and Computer Engineering — Aug 2026
M.S. in Electrical and Computer Engineering — Dec 2025

University of Pennsylvania, PA, USA
M.S.E. in Nanotechnology — May 2020

Amity University, Noida, India
B.Tech. in Nanotechnology — May 2018

Selected Awards

- **Hayes Forum Poster Competition,** 3rd Place (2025)
- **Best Student Paper Award,** Honorable mention at the Electronic Materials Conference (2024)